$$C_\phi^{(22)\;\;(22)\;\;(22)} \to$$

$$\frac{1}{16\pi^2}\Big(\frac{1}{18} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,0}[m_2] + \frac{1}{12} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,-1}[m_2] - \frac{4}{45} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{5,-2}[m_2] -$$

$$\frac{1}{216}\sum_p g_1^6 \mathrm{LF}_{3,0}\left[m_{\tilde{d}}^{p}\right] + \frac{1}{12} g_1^4 \overline{y_d}^{pr} y_d^{pr} \mathrm{LF}_{3,0}\left[m_{\tilde{d}}^{r}\right] -$$

$$\frac{1}{24}\sum_p g_1^6 \mathrm{LF}_{3,0}\left[m_{\tilde{e}}^{p}\right] + \frac{1}{4} g_1^4 \overline{y_e}^{pr} y_e^{pr} \mathrm{LF}_{3,0}\left[m_{\tilde{e}}^{r}\right] +$$

$$\frac{1}{288}\left(18\,\overline{y_e}^{pr} y_e^{pr}\left(g_1^4+2 g_1^2 g_2^2+5 g_2^4\right) + \sum_p \left(3 g_1^6 - g_1^2 g_2^4 - 10 g_2^6\right)\right) \mathrm{LF}_{3,0}\left[m_{\tilde{l}}^{p}\right] +$$

$$\frac{1}{96} g_2^2 \left(g_1^2+g_2^2\right)\left(-24\,\overline{y_e}^{pr} y_e^{pr} + \sum_p g_2^2\right) \mathrm{LF}_{4,-1}\left[m_{\tilde{l}}^{p}\right] + \frac{1}{45}\sum_p g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{5,-2}\left[m_{\tilde{l}}^{p}\right] +$$

$$\frac{1}{864}\left(18\,\overline{y_d}^{pr} y_d^{pr}\left(g_1^4+42 g_1^2 g_2^2+45 g_2^4\right) - \sum_p \left(g_1^6 + 117 g_1^2 g_2^4 + 90 g_2^6\right)\right) \mathrm{LF}_{3,0}\left[m_{\tilde{q}}^{p}\right] +$$

$$\frac{1}{32} g_2^2 \left(g_1^2+g_2^2\right)\left(-24\,\overline{y_d}^{pr} y_d^{pr} + \sum_p g_2^2\right) \mathrm{LF}_{4,-1}\left[m_{\tilde{q}}^{p}\right] + \frac{1}{15}\sum_p g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{5,-2}\left[m_{\tilde{q}}^{p}\right] +$$

$$\frac{1}{27}\sum_p g_1^6 \mathrm{LF}_{3,0}\left[m_{\tilde{u}}^{p}\right] + \frac{1}{36} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,0}[\tilde{\mu}] + \frac{1}{24} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,-1}[\tilde{\mu}] -$$

$$\frac{2}{45} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{5,-2}[\tilde{\mu}] - \frac{1}{12} g_1^6 \mathrm{LF}_{3,3,-3}[m_1, \tilde{\mu}] - \frac{1}{4} g_1^6 m_1^2 \mathrm{LF}_{3,3,-2}[m_1, \tilde{\mu}] -$$

$$\frac{1}{3} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,1,0}[m_2, \tilde{\mu}] - \frac{17}{24} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,-1}[m_2, \tilde{\mu}] +$$

$$\frac{1}{6} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,-1}[m_2, \tilde{\mu}] + \frac{7}{6} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-2}[m_2, \tilde{\mu}] -$$

$$\frac{3}{4} g_2^4 m_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] - \frac{1}{4} g_2^4 \left(2 g_1^2+5 g_2^2\right) \mathrm{LF}_{3,3,-3}[m_2, \tilde{\mu}] +$$

$$\frac{1}{4} g_2^4 m_2^2 \left(2 g_1^2+g_2^2\right) \mathrm{LF}_{3,3,-2}[m_2, \tilde{\mu}] - \frac{1}{2} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,2,-3}[m_2, \tilde{\mu}] +$$

$$\frac{1}{2} g_2^4 m_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,2,-2}[m_2, \tilde{\mu}] - \frac{1}{2} g_1^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \mathrm{LF}_{2,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}\right] +$$

$$\frac{1}{12} g_1^2 \overline{a_d}^{pr} a_d^{pr} \left(2 g_1^2+3 g_2^2\right) \mathrm{LF}_{2,2,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] + \frac{1}{12} g_1^4 \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] -$$

$$\frac{1}{2} g_1^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \mathrm{LF}_{2,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}\right] + \frac{1}{4} g_1^2 g_2^2 \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{2,2,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] +$$

$$\frac{1}{4} g_1^4 \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] + \frac{1}{4} g_2^4 \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] -$$

$$\frac{1}{8} g_1^2 \overline{a_e}^{pr} a_e^{pr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] - \frac{1}{8} g_2^2 \overline{a_e}^{pr} a_e^{pr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] +$$

$$\frac{1}{24} \overline{a_e}^{pr} a_e^{pr} \left(3 g_1^2-g_2^2\right)\left(g_1^2+g_2^2\right) \mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] -$$

$$\frac{1}{2} g_2^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \mathrm{LF}_{2,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] + \frac{1}{4} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] +$$

$$\frac{1}{12} \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\left(2 g_1^2+3 g_2^2\right)^2 - \frac{1}{8} g_1^2 \overline{a_d}^{pr} a_d^{pr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] -$$

$$\frac{3}{8} \overline{a_d}^{pr} a_d^{pr} \left(g_1^2+g_2^2\right)\left(2 g_1^2+g_2^2\right) \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] + \frac{1}{8} \overline{a_d}^{pr} a_d^{pr} \left(3 g_1^2-g_2^2\right)$$

$$\left(g_1^2+g_2^2\right) \mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] - \frac{1}{2} \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \left(2 g_1^2+3 g_2^2\right) \mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{4} \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] - \frac{1}{48} \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \left(g_1^4 - 18 g_1^2 g_2^2 - 3 g_2^4\right)$$

$$\mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{1}{48} \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \left(g_1^4 - 42 g_1^2 g_2^2 - 27 g_2^4\right) \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] +$$

$$\frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \left(g_1^2+g_2^2\right)$$

$$\mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{1}{48} \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \left(g_1^4 - 18 g_1^2 g_2^2 - 3 g_2^4\right) \mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] -$$

$$\frac{1}{16} \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\left(g_1^2+g_2^2\right)^2 + \frac{3}{16} \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\left(g_1^2+g_2^2\right)^2 +$$

$$\frac{1}{8} \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} \left(3 g_1^2-g_2^2\right)\left(g_1^2+g_2^2\right) \mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] - \frac{1}{8} g_1^2 \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_1]\left(g_1^2+g_2^2\right)^2 +$$

$$\frac{3}{16} g_1^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_1] + \frac{3}{16} g_1^4 m_1^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] +$$

$$\frac{1}{24} g_1^2 \left(g_1^2+g_2^2\right)\left(6 g_1^2+g_2^2\right) \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1] - \frac{1}{8} g_1^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,2,-3}[\tilde{\mu}, m_1] -$$

$$\frac{1}{8} g_1^4 m_1^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{24}\left(-3 g_1^6 - 2 g_1^4 g_2^2 + g_1^2 g_2^4\right) \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1] +$$

$$\frac{1}{24}\left(-9 g_1^4 g_2^2 - 2 g_1^2 g_2^4 + 7 g_2^6\right) \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2] +$$

$$\frac{85}{48} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_2] + \frac{3}{16} g_2^4 m_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] +$$

$$\frac{3}{8}\left(2 g_1^4 g_2^2 + g_1^2 g_2^4 - g_2^6\right) \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2] - \frac{7}{8} g_2^4 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,2,-3}[\tilde{\mu}, m_2] -$$

$$\frac{1}{8} g_2^4 m_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{8}\left(-3 g_1^4 g_2^2 - 2 g_1^2 g_2^4 + g_2^6\right) \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2] +$$

$$g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] + \frac{3}{4} m_1 m_2 g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] -$$

$$\frac{1}{2} g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] - \frac{1}{2} m_1 m_2 g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] +$$

$$\overline{y_d}^{pr} \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_d^{ur} \mathrm{LF}_{1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}\right] - \frac{1}{2} g_1^2 \overline{y_d}^{st} y_d^{sr} \overline{a_d}^{pr} a_d^{pt}$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}\right] - \frac{1}{2} g_1^2 \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] -$$

$$\frac{1}{2} g_1^2 \overline{y_d}^{st} y_d^{sr} \overline{a_d}^{pr} a_d^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] + \frac{1}{3} \overline{y_e}^{pr} \overline{y_e}^{st} \overline{y_e}^{uv} y_e^{pt} y_e^{sv} y_e^{ur}$$

$$\mathrm{LF}_{1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}\right] - \frac{1}{2} g_1^2 \overline{y_e}^{st} y_e^{sr} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{2} g_1^2 \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] - \frac{1}{2} g_1^2 \overline{y_e}^{st} y_e^{sr} \overline{a_e}^{pr} a_e^{pt}$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] - \frac{1}{2} g_2^2 \overline{y_e}^{st} y_e^{sr} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}\right] +$$

$$\frac{1}{4} \overline{y_e}^{st} y_e^{sr} \overline{a_e}^{pr} a_e^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}\right] - \frac{1}{2} g_2^2 \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr}$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}\right] + \frac{1}{4} \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}\right] +$$

$$\frac{1}{3} \overline{y_e}^{pr} \overline{y_e}^{st} \overline{y_e}^{uv} y_e^{pt} y_e^{sv} y_e^{ur} \mathrm{LF}_{1,1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}\right] - \frac{1}{2} g_2^2 \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr}$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] + \frac{1}{4} \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{2} \overline{y_d}^{st} y_d^{sr} \overline{a_d}^{pr} a_d^{pt} \left(2 g_1^2+3 g_2^2\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}\right] +$$

$$\frac{3}{4} \overline{y_d}^{st} y_d^{sr} \overline{a_d}^{pr} a_d^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}\right] -$$

$$\frac{1}{2} \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \left(2 g_1^2+3 g_2^2\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{4} \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] +$$

$$\overline{y_d}^{pr} \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_d^{ur} \mathrm{LF}_{1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}\right] +$$

$$\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] -$$

$$\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] -$$

$$\frac{3}{8} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}\right] -$$

$$\frac{1}{2} \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \left(2 g_1^2+3 g_2^2\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] +$$

$$\frac{3}{4} \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] +$$

$$\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] -$$

$$\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] -$$

$$\frac{3}{8} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}\right] -$$

$$\frac{3}{8} g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] + \frac{7}{8} g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] +$$

$$\frac{3}{8} m_1 m_2 g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] - \frac{1}{4} g_1^4 g_2^2 \mathrm{LF}_{3,2,1,-3}[\tilde{\mu}, m_1, m_2] -$$

$$\frac{1}{4} m_1 g_1^4 g_2^2 (m_1 + 2 m_2) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_1, m_2] - \frac{1}{2} g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] -$$

$$\frac{1}{4} m_1 m_2 g_1^2 g_2^2 \left(g_1^2+g_2^2\right) \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \frac{1}{4} g_1^2 g_2^2 \left(2 g_1^2+3 g_2^2\right) \mathrm{LF}_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] -$$

$$\frac{1}{4} m_2 g_1^2 g_2^2 \left(2 m_1 g_1^2 + g_2^2 (4 m_1 + m_2)\right) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] +$$

$$3\,\overline{y_d}^{st} \overline{y_d}^{uv} y_d^{sv} y_d^{ur} \overline{a_d}^{pr} a_d^{pt} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}\right] -$$

$$\frac{1}{2} g_1^2 \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] +$$

$$3\,\overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pt} y_d^{ur} \overline{a_d}^{pr} a_d^{sv} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] +$$

$$3\,\overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pv} y_d^{ut} \overline{a_d}^{pr} a_d^{sr} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}\right] +$$

$$\overline{y_e}^{st} \overline{y_e}^{uv} y_e^{sv} y_e^{ur} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{2} g_1^2 \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] +$$

$$\overline{y_e}^{st} \overline{y_e}^{uv} y_e^{pt} y_e^{ur} \overline{a_e}^{pr} a_e^{sv} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] +$$

$$\overline{y_e}^{st} \overline{y_e}^{uv} y_e^{pv} y_e^{ut} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}\right] -$$

$$\frac{1}{2} g_2^2 \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}\right] +$$

$$\frac{1}{4} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}\right] -$$

$$\frac{1}{2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sr} \left(2 g_1^2+3 g_2^2\right) \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{4} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{4} \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} a_d^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}\right] -$$

$$g_1^2 \tilde{\mu}^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{4} \tilde{\mu}^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{8} \tilde{\mu}^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{8} \tilde{\mu}^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] +$$

$$\frac{1}{4} \tilde{\mu}^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \left(g_1^2-3 g_2^2\right) \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{4} \tilde{\mu}^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{8} \tilde{\mu}^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \left(g_1^2+g_2^2\right) \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] +$$

$$3\,\overline{y_d}^{uv} y_d^{ur} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sv} \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] +$$

$$3\,\overline{y_d}^{uv} y_d^{sv} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{ur} \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}\right] +$$

$$\overline{y_e}^{uv} y_e^{ur} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sv} \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] +$$

$$\overline{y_e}^{uv} y_e^{sv} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{ur} \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}\right] +$$

$$\overline{a_d}^{pr} \overline{a_d}^{st} \overline{a_d}^{uv} a_d^{pt} a_d^{sv} a_d^{ur} \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}\right] +$$

$$\frac{1}{3} \overline{a_e}^{pr} \overline{a_e}^{st} \overline{a_e}^{uv} a_e^{pt} a_e^{sv} a_e^{ur} \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}\right] +$$

$$\tilde{\mu}^6 \overline{y_u}^{pr} \overline{y_u}^{st} \overline{y_u}^{uv} y_u^{pt} y_u^{sv} y_u^{ur} \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}\right]\Big)$$